

Use Case Flows

Attendance Management System

UC-01: Login (Role-based Access)

Related Requirement: FR-01

Actors: Administrator, Teacher, Student

Preconditions:

User has a valid account registered in the system.

Main Flow

1. User navigates to the login page.
2. User enters their username/ID and password.
3. System sends credentials to the Security/Auth Service for verification.
4. Security/Auth Service validates the credentials and returns the user's role.
5. System grants access and redirects the user to the dashboard corresponding to their role.

Alternate / Exception Flow

3a. Invalid credentials entered: system displays an error message and prompts re-entry.

4a. Account locked or disabled: system denies access and directs the user to contact the administrator.

Postconditions

User is authenticated and has role-appropriate access to system functions.

UC-02: Select Class / Subject

Related Requirement: FR-02

Actors: Teacher

Preconditions:

Teacher is logged in.

Main Flow

1. Teacher navigates to the Attendance module.
2. System displays the list of classes and subjects assigned to the Teacher.
3. Teacher selects a class and subject.
4. Teacher selects the date for attendance.
5. System loads the student roster for the selected class, subject, and date.

Alternate / Exception Flow

2a. No classes assigned to the Teacher: system displays a message indicating no classes are available.

Postconditions

The student roster is ready for attendance marking.

UC-03: Mark Attendance (Present / Absent / Late)

Related Requirement: FR-02

Actors: Teacher

Preconditions:

Class and subject have been selected (UC-02).

Main Flow

1. System displays the roster of students for the selected class, subject, and date.
2. Teacher marks each student's status as Present, Absent, or Late.
3. Teacher reviews the marked attendance for accuracy.
4. Teacher submits the attendance.
5. System proceeds to save the record.

Alternate / Exception Flow

- 2a.** A student is left unmarked: system prompts the teacher to complete all entries before submission.

Postconditions

Attendance statuses are captured and ready to be saved.

UC-04: Save / Update Attendance Records

Related Requirement: FR-03

Actors: Teacher

Preconditions:

Attendance has been marked (UC-03).

Main Flow

1. System receives the submitted attendance data.
2. System checks for an existing record for the same student, subject, and date.
3. If no duplicate exists, system saves the new record to the Attendance Database.
4. If a record already exists, system allows the authorized teacher to update it instead of creating a duplicate.
5. System confirms the successful save or update to the Teacher.
6. System triggers recalculation of the attendance percentage.

Alternate / Exception Flow

- 3a.** Save operation fails (e.g., connection issue): system displays an error and prompts the teacher to retry, without loss of data already entered.

Postconditions

The attendance record is persisted accurately with no duplicates.

UC-05: View Attendance Records

Related Requirement: FR-05

Actors: Teacher, Student

Preconditions:

User is logged in; attendance records exist for the relevant class or subject.

Main Flow

1. User navigates to the Attendance Records section.
2. User selects filters, such as subject, date, date range, or student, depending on role.
3. System retrieves matching records from the Attendance Database.
4. System displays the attendance records along with the calculated attendance percentage.

Alternate / Exception Flow

3a. No records found for the selected filters: system displays a message indicating no data is available.

Postconditions

The user views accurate, up-to-date attendance information limited to what their role permits.

UC-06: Calculate Attendance Percentage

Related Requirement: FR-04

Actors: System (automatic); results viewed by Teacher and Student

Preconditions:

Attendance records exist for the student and subject.

Main Flow

1. System retrieves all attendance records for the student and subject from the Attendance Database.
2. System counts total classes held and classes attended.
3. System calculates the attendance percentage.
4. System displays the percentage alongside the attendance record view or report.

Alternate / Exception Flow

2a. No attendance records exist yet: system displays 0% or marks the value as not available.

Postconditions

An accurate, current attendance percentage is available for display.

UC-07: Generate Attendance Reports

Related Requirement: FR-06

Actors: Teacher, Administrator

Preconditions:

Attendance records exist for the selected scope.

Main Flow

1. User navigates to the Reports section.
2. User selects the report scope: by student, class, subject, or date range.
3. System retrieves attendance records matching the scope.
4. System compiles the data into a report through the Report Engine.
5. System displays or exports the report to the user.

Alternate / Exception Flow

3a. No data available for the selected scope: system notifies the user that no report can be generated.

Postconditions

A complete and accurate attendance report is produced within acceptable performance limits.

Cross-Cutting Non-Functional Notes

- Security (NFR-01): enforced at Login and View Attendance Records; role-based access ensures students only see their own data, and data is protected in transit.
- Performance (NFR-02): applies to View Attendance Records and Generate Reports; page and report loads should complete within 3 seconds under normal conditions.
- Reliability (NFR-03): applies to Save/Update Attendance Records; saved records must not be lost, and regular backups must be performed.